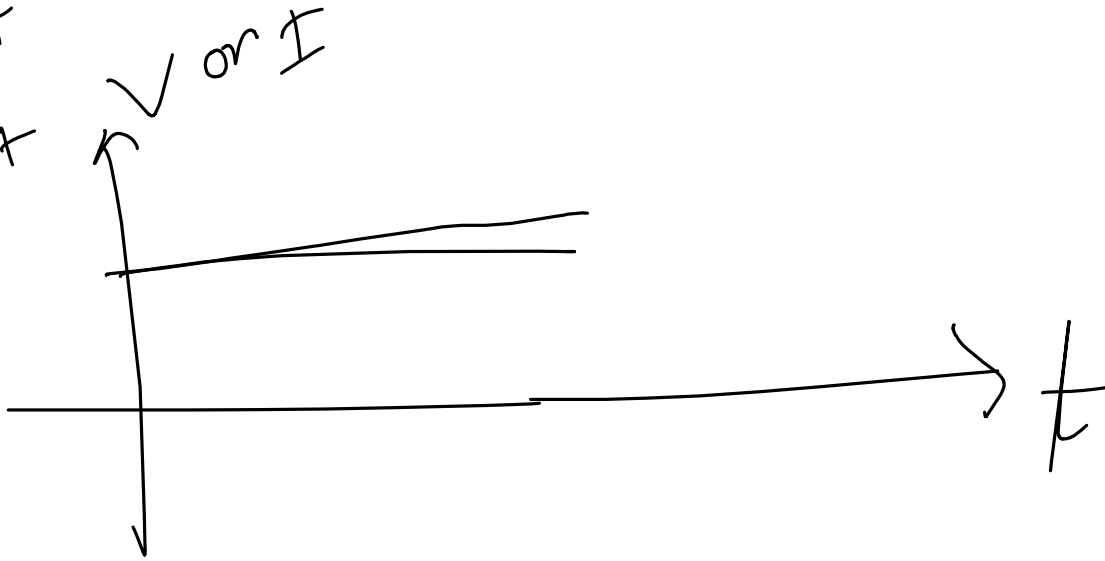


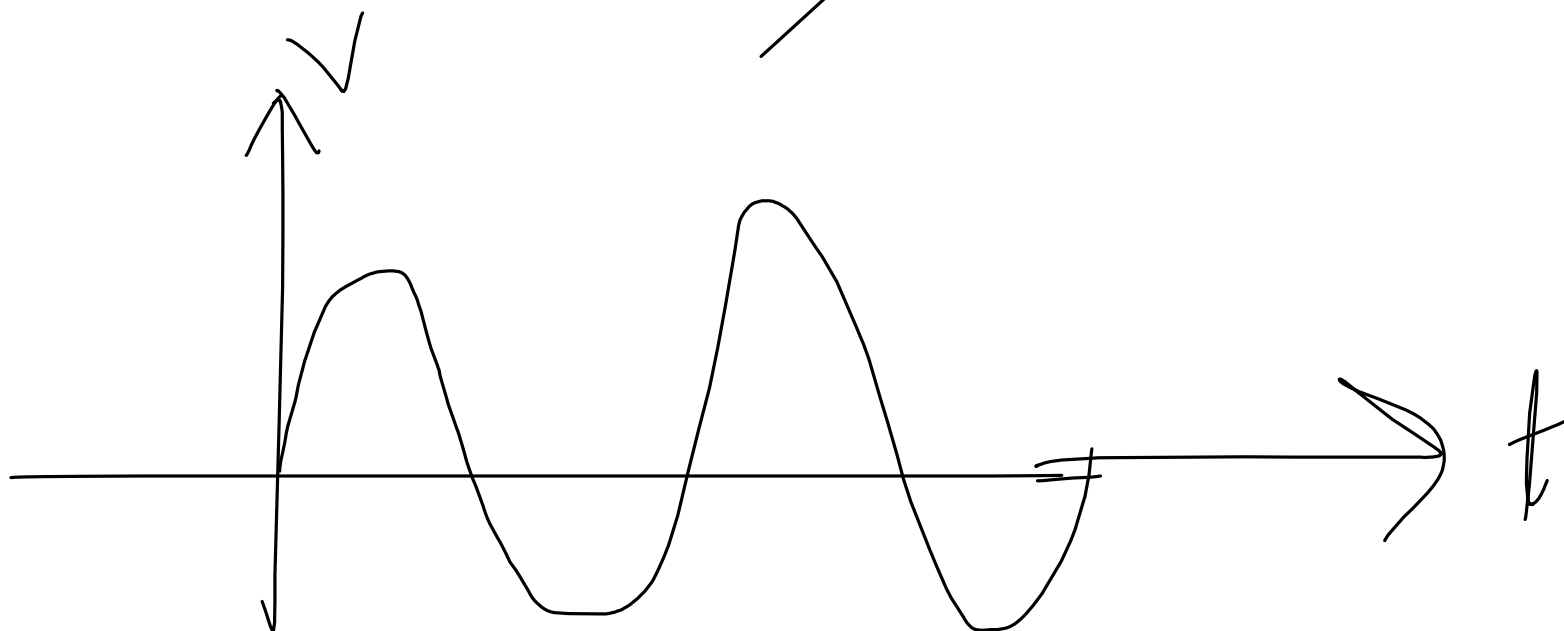
AC CKT analysis

voltage +
current



AC

→ Sine wave.



AC

sine / cos



sinusoid

$$V = A \sin(\omega t + \phi)$$

Amplitude

angular freq

phase angle / initial angle

$$f = \frac{1}{T}$$

$$\omega = \frac{2\pi f}{T}$$

$$= \frac{2\pi}{\omega}$$

condⁿ of periodicity

$$f(t+T) = f(t)$$

$f(t)$

✓

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

$$A \sin(\omega t + \phi)$$

$\downarrow$
A

$$A \cos(\omega t + \delta)$$

$\downarrow$
sin

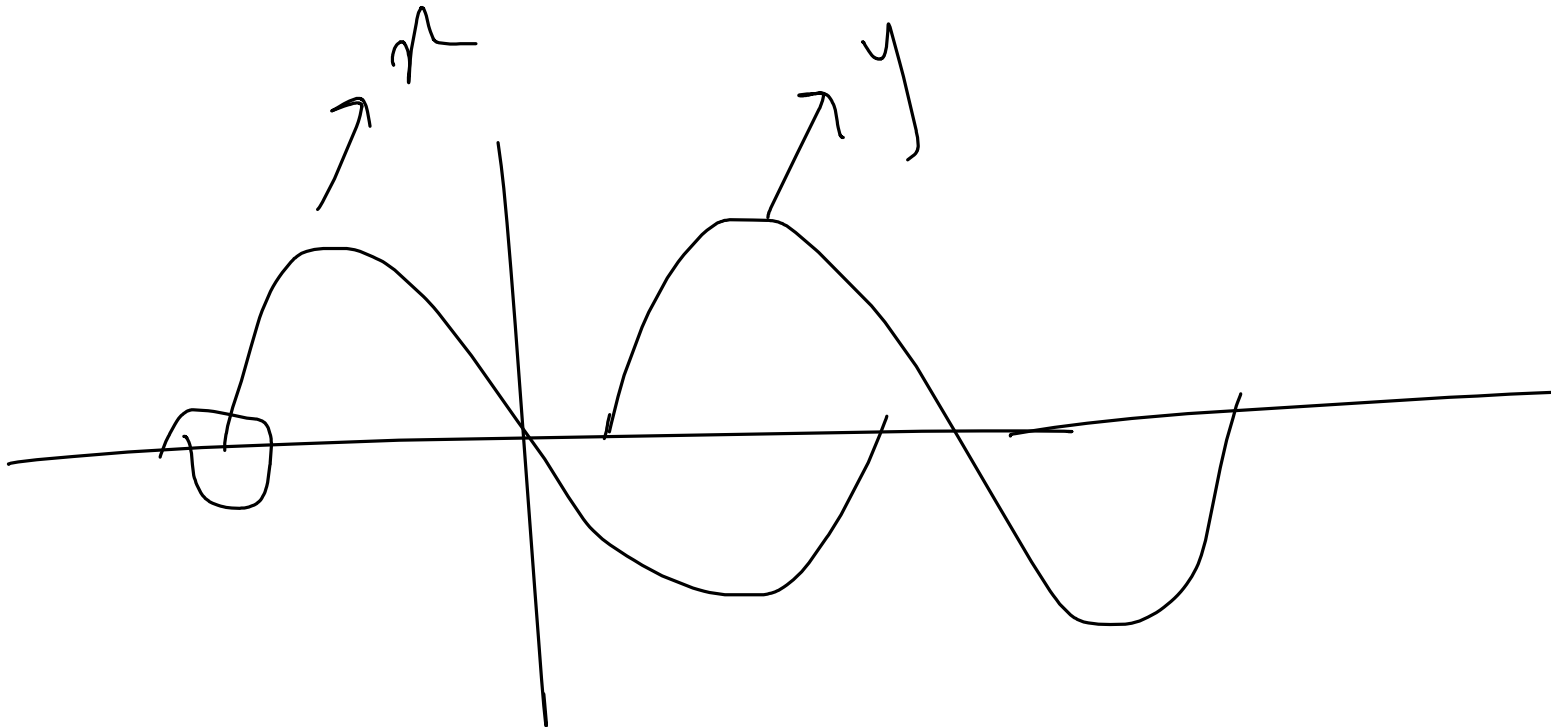
Given the sinusoid $30 \sin(4\pi t - 75^\circ)$, calculate its amplitude, phase, angular frequency, period, and frequency.

Handwritten annotations and formulas:

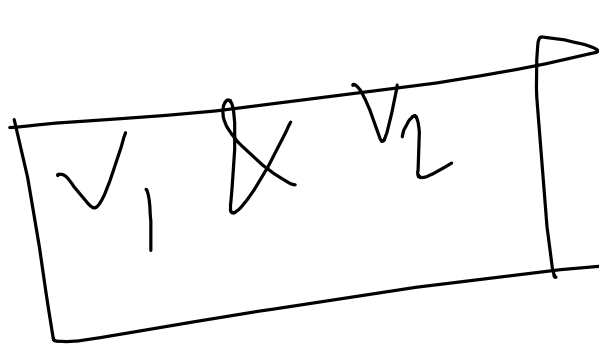
- Amplitude: 30
- Phase: -75°
- Angular frequency: 4π
- Period: $T = \frac{2\pi}{\omega}$
- Frequency: $f = \frac{1}{T}$

Additional handwritten notes: ωk and ω with arrows pointing to the angular frequency term in the original expression.

Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.



Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.



$$\begin{aligned} \sin(\omega t \pm 180^\circ) &= -\sin \omega t \\ \cos(\omega t \pm 180^\circ) &= -\cos \omega t \end{aligned}$$

~~1~~ All amplitude sign must be
 same. ✓
 same trigonometric variation.

$$\begin{aligned} \sin(\omega t \pm 90^\circ) &= \pm \cos \omega t \\ \cos(\omega t \pm 90^\circ) &= \mp \sin \omega t \end{aligned}$$

Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

$$\begin{aligned}\sin(\omega t \pm 180^\circ) &= -\sin \omega t \\ \cos(\omega t \pm 180^\circ) &= -\cos \omega t\end{aligned}$$

$$\begin{aligned}v_1 &= -10 \cos(\omega t + 50^\circ) \\ &= 10 \cos(\omega t + 50^\circ + 180^\circ) \\ &= 10 \cos(\omega t + 230^\circ)\end{aligned}$$

$$V_2 = 12 \sin(\omega t - 10^\circ)$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

$$= 12 \cos(\omega t - 10^\circ - 90^\circ)$$

$$= 12 \cos(\omega t - 100^\circ)$$

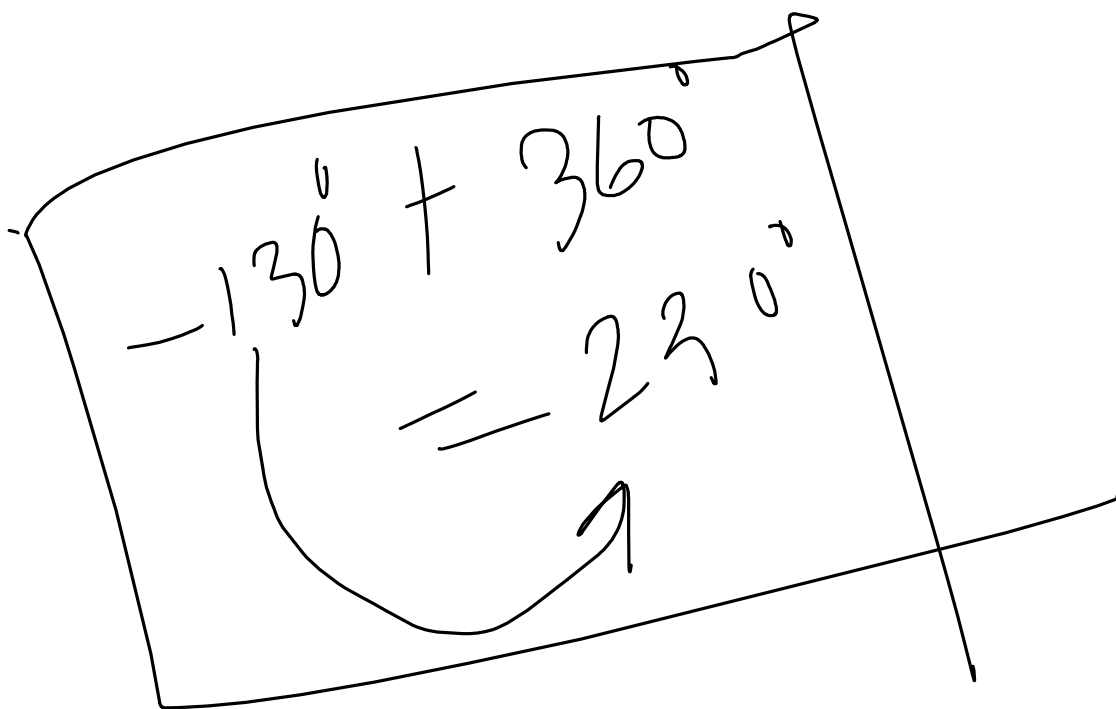
V_1 alternate

$$\begin{aligned}\sin(\omega t \pm 180^\circ) &= -\sin \omega t \\ \cos(\omega t \pm 180^\circ) &= -\cos \omega t\end{aligned}$$

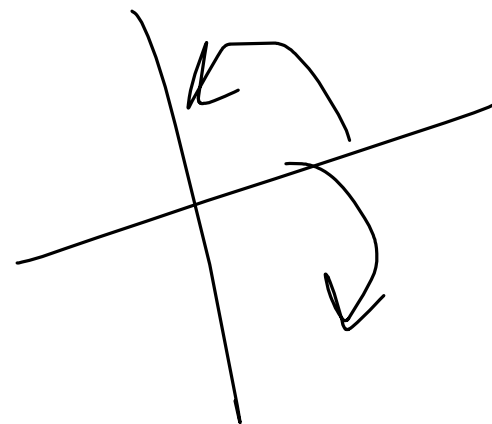
$$\begin{aligned}V_1 &= -10 \cos(\omega t + 5^\circ) \\ &= -10 \cos(\omega t + 5^\circ - 180^\circ) \\ &= -10 \cos(\omega t - 130^\circ)\end{aligned}$$

$$\begin{array}{|c|} \hline -180^\circ \\ \hline \end{array} \quad \begin{array}{|c|} \hline -130^\circ \\ \hline \end{array}$$

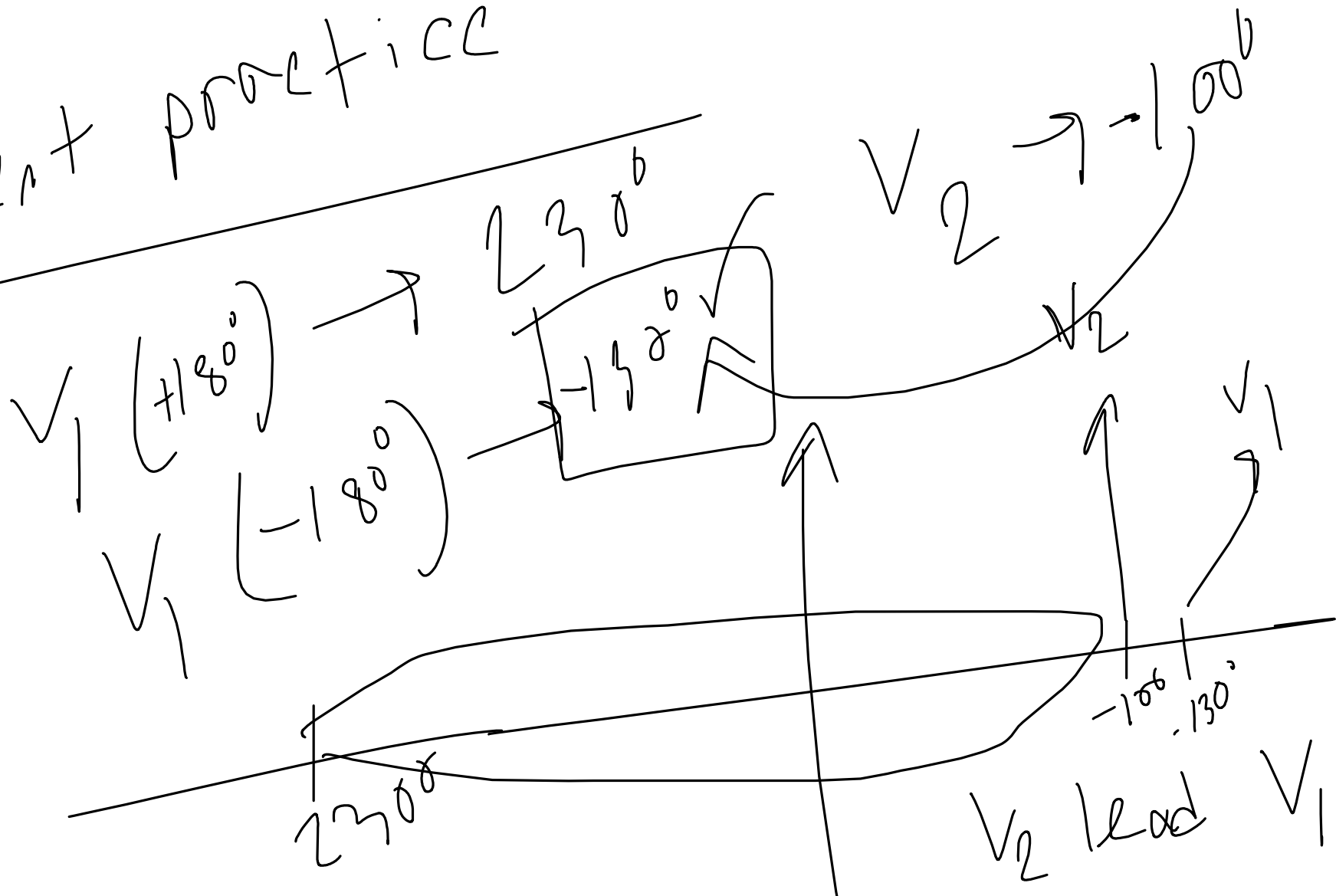
$$+180^\circ \longrightarrow \begin{array}{|c|} \hline 230^\circ \\ \hline \end{array}$$



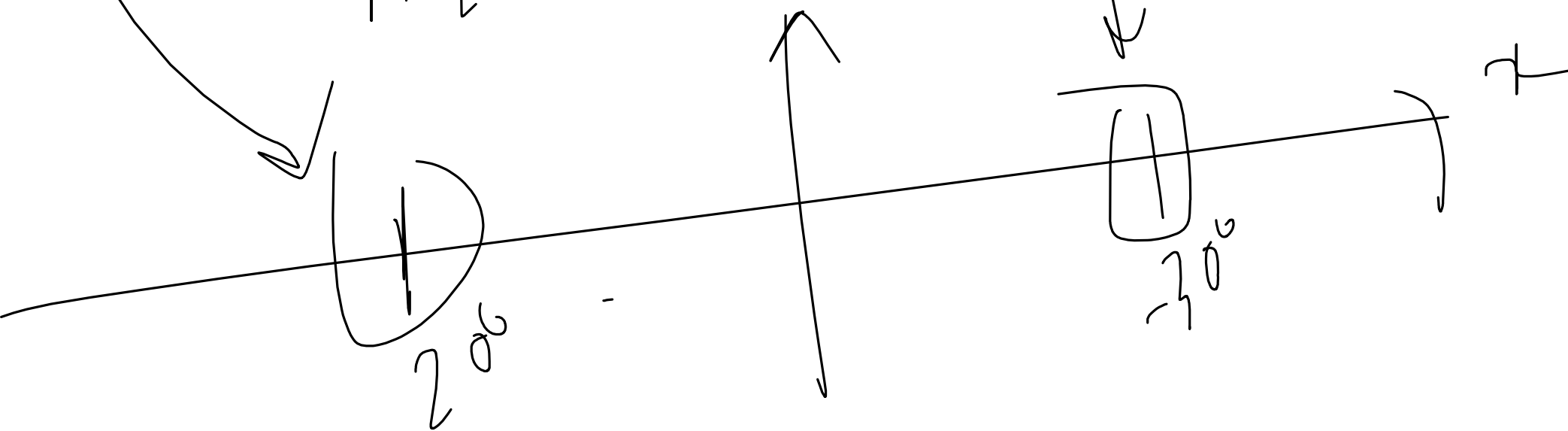
8



best practice



$$A_1 \sin(\omega t + 20^\circ)$$
$$A_2 \sin(\omega t - 20^\circ)$$



Find the phase angle between

$$i_1 = -4 \sin(377t + 55^\circ) \quad \text{and} \quad i_2 = 5 \cos(377t - 65^\circ)$$

Does i_1 lead or lag i_2 ?

$$\begin{aligned} \sin(\omega t \pm 180^\circ) &= -\sin \omega t \\ \cos(\omega t \pm 180^\circ) &= -\cos \omega t \end{aligned}$$

Step 1

$$\begin{aligned} i_1 &= -4 \sin(377t + 55^\circ) \\ &= 4 \sin(377t + 55^\circ + 180^\circ) \\ &= 4 \sin(377t + 235^\circ) \end{aligned}$$

Step-2

$i_1 =$

$$\begin{aligned} &= 4 \sin(377t + 235^\circ) \\ &\quad - 4 \cos(377t + 235^\circ - 90^\circ) \\ &= 4 \cos(377t + 145^\circ) \\ \therefore i_1 &= 4 \cos(377t - 65^\circ) \\ i_2 &= 5 \cos(377t - 65^\circ) \end{aligned}$$

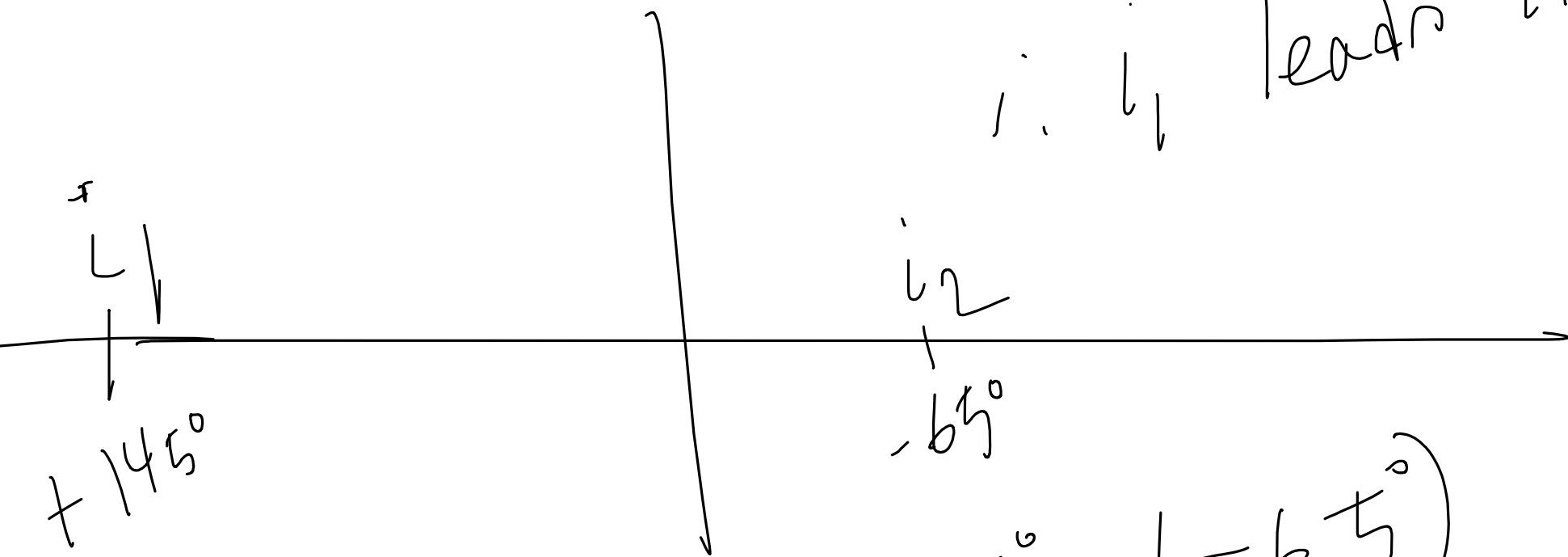
i_1

i_2

i_2

$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$
 $\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$

i_1 leads i_2



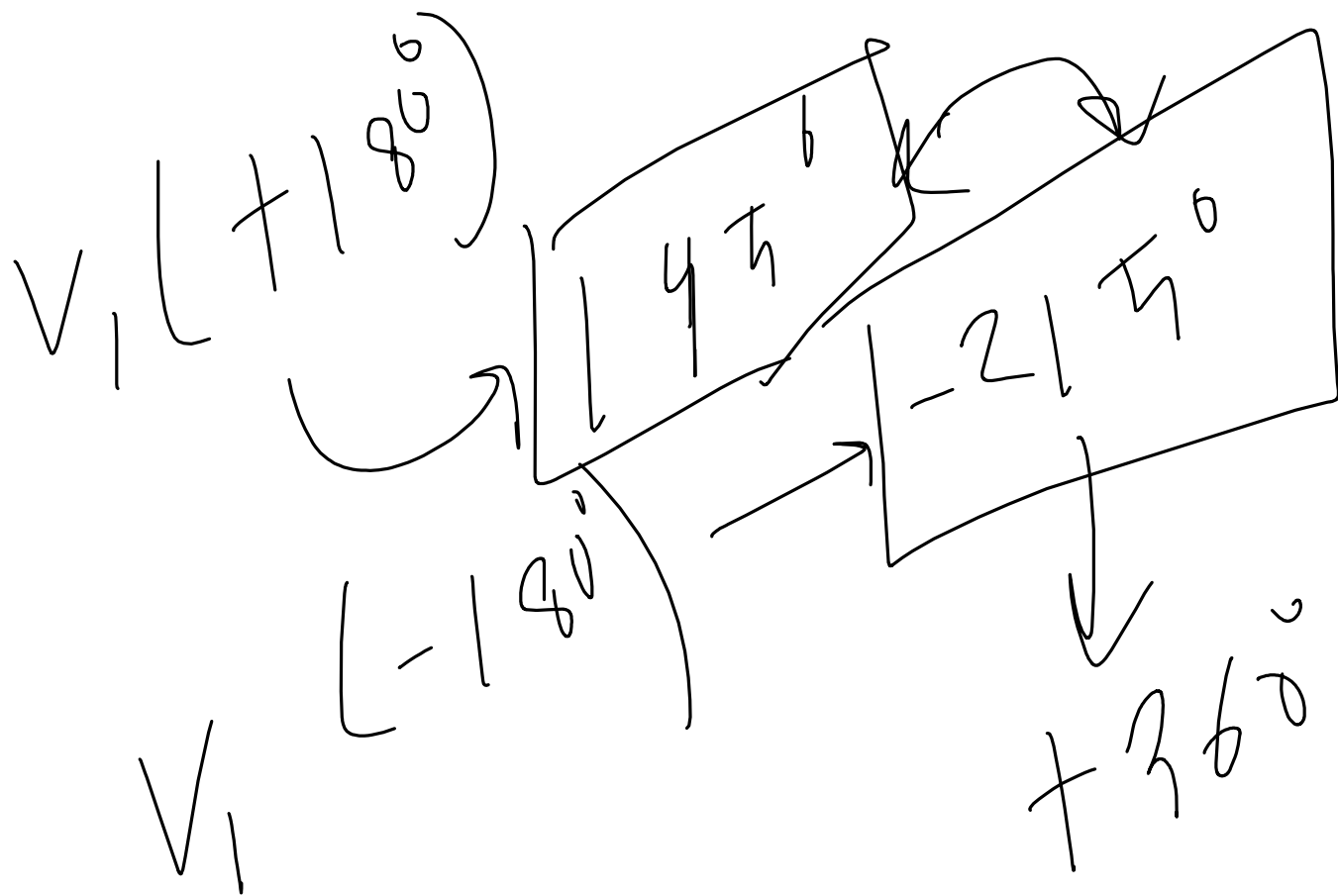
$$\Delta\phi = 145^\circ - (-65^\circ)$$
$$= 210^\circ$$

$$\begin{aligned}
 \frac{5 \times 10^{-1}}{4} &= -4 \sin(377t + 75^\circ) \\
 &= 4 \sin(377t + 55^\circ - 180^\circ) \\
 &= 4 \sin(377t - 125^\circ) \\
 &\rightarrow
 \end{aligned}$$

Step 2

$$\begin{aligned} i_1 &= 4 \sin(377t - 125^\circ) \\ &= 4 \cos(377t - 125^\circ - 90^\circ) \\ &= 4 \cos(377t - 215^\circ) \end{aligned}$$

$$\Delta \phi = 150^\circ$$



$$+360^\circ = 144^\circ$$

